ADRIAN EGEA

SENIOR C++ SOFTWARE ENGINEER | UNREAL GAME DEVELOPER

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PROFESSIONAL SUMMARY

Software Engineer with **9+ years of experience** specializing in software architecture, gameplay and systems programming, performance optimization, cross-platform development, and Modern C++ (17/20/23). Graphics API enthusiast with a custom **Vulkan 3D Game Engine**. I've taken part in multiple **AA/AAA Unreal Engine** projects across all production stages (pre-release, mid-production, and greenfield), such as **Lords of the Fallen** (2023, PS5, XSX, PC), **Gothic Remake** (2026, PS5, XSX, PC), and **Remothered Tormented Fathers Remaster** / **Remothered Broken Porcelain Remaster** (2026, PS5, XSX, Switch 2, PC). I bring a strong engineering mindset, adaptability, and a hands-on approach to solving complex technical challenges.

TECHNICAL SKILLS

- **Programming Languages**: C++(17/20/23), C, Python, JavaScript, HTML, CSS
- **Programming Concepts**: Memory Management, Smart Pointers, Multithreading, Data Structures, ECS
- **Software Engineering**: Software Architecture, Object-Oriented, Data-Driven, Pair-Programming, Agile/Scrum, TDD
- **Graphics Programming**: Vulkan, OpenGL, WebGL, GLSL, 3D, 2D, Maths, GPU optimization, PBR
- **Build and Automation**: Linux, CMake, GCC, Clang, Jenkins, TeamCity, Bazel, Perforce, Git, Gitea, Docker, Portainer
- **Frameworks**: STL, Unreal Engine 5, Blueprints, Replication, GAS, UBT, PlayStation SDK, Switch SDK, Xbox SDK
- **Tools**: Visual Studio, VSCode, Rider, Jira, Unreal Automation, GoogleTest
- **Debug and Profiling**: GDB, Valgrind, Address Sanitizer, Vulkan Validation Layers, RenderDoc, NVIDIA Nsight, Tracy, Dear ImGui, Unreal Profiler

PROFESSIONAL EXPERIENCE

Mobile C++ Programmer | King (Contract) Aug 2026 - Present (3m)

- Contribute to the implementation and maintenance of C++ production software in a mobile development environment, Android and iOS.
- Implemented UI popups and server error handling for a new mini-game using King's internal framework, with unit tests in GoogleTest and automated builds via Jenkins and Bazel.

Principal UE5 C++ Programmer | Radical Dreams Entertainment (Contract) Feb 2026 - Aug 2026 (7m)

- Developed cross-platform **Remaster** ports for **Remothered Tormented Fathers** and **Remothered Broken Porcelain** (PC, PS5, XSX, Switch 2).
- Contributed to **porting** existing software to Unreal Engine 5 and resolving platform-specific issues.
- Implemented and maintained gameplay and system-level functionality, UI components, and cinematic systems.
- Set up **CI/CD** infrastructure using **TeamCity** and **Docker** to support automated builds and development workflows.

Unreal Engine 5 C++ Programmer | GRIP Studios (Contract) Apr 2025 - Jan 2026 (10m)

- Designed and implemented a **Modding system** based in Mod.io, supporting PC and console platforms for unannounced **Simulation** project (PC, PS5, XSX).
- Developed gameplay systems and refactored existing **Unreal Mutable plugin** architecture to support runtime generation of characters and vehicles.
- Built debugging and visualization tools to improve development workflows and facilitate issue diagnosis.

Unreal Engine 5 C++ Programmer | Underdog Studio (CI Games) Mar 2024 - Oct 2024 (8m)

- Developed a Quest management system for an unannounced multiplayer **Survival** project (PS5, XSX, PC) and Quest system state replication to support cooperative multiplayer gameplay.

Unreal Engine 5 C++ Programmer | HexWorks (CI Games) Aug 2021 - Mar 2024 (2y 8m)

- Designed, implemented, and maintained core software systems for **Lords of the Fallen** (2023, PS5, XSX, PC), including quest management, save-game persistence, and data serialization, supporting dozens of quests and their variants.
- Developed gameplay systems involving inventory management, character statistics, Niagara VFX and Wwise audio integration.
- Contributed to multiplayer networking, state replication, and gameplay ability systems (GAS).
- Investigated and resolved critical software defects during development and post-release maintenance, including patches and DLCs.
- Optimized multithreaded asynchronous asset and data loading, and improved gameplay performance using asynchronous Ability Tasks (GAS).

Unreal Engine 4 C++ Programmer | Alkimia Interactive (THQ Nordic) May 2020 - Jul 2021 (1y 3m)

- Worked on **Gothic Remake** (2026; PS5, XSX, PC).
- Developed character and creature movement systems and animation-related functionality for quadrupeds.
- Migrated existing ALS Plugin implementation to C++ to improve code performance and maintainability.
- Implemented and debugged AI functionality for small creature behaviors (birds, rats, fish…).
- Contributed to movement systems and gameplay abilities within an established software architecture.

Lead Unreal Engine 4 C++ Programmer | Limitless Apr 2018 - Mar 2020 (2y)

- Worked on a cancelled, unannounced **MOBA** project (PC).
- Led a 5 developer programming team, coordinating implementation tasks and technical decisions.
- Defined software architecture and implemented networking functionality for multiplayer.
- Developed automated CI/CD build pipelines with Jenkins and Git, plus internal tools and Git automation scripts to improve art team workflows.

Graphics Developer | Cl3ver May 2017 - Apr 2018 (1y)

- Developed C++ and OpenGL functionality for an **Oculus Rift** application. Implemented 3D data importer plugins for **Rhino** and **SketchUp**.

Full-Stack / Android Developer | Tretanto Feb 2015 - Apr 2016 (1y 3m)

- Developed Web, Android applications and social networking features using JS, jQuery, AngularJS, Android, PHP, Symfony2, Doctrine, and MySQL.

SIDE PROJECTS

C++ / OpenGL / Vulkan 3D Game Engine Nov 2017 - Present (9y)

- Custom Vulkan 3D engine built for graphics and systems learning and research. Made with C++, CMake, Python, Vulkan, OpenGL 4.x, GLSL, SPIRV. GitHub Repo
- **Features**: 2D, 3D, Hybrid ECS, Automatic Batching, Instanced Rendering, Data Driven Shaders Builder, Texture Atlases, Animation Spritesheets, Built-In Custom UI, Scene management, Perspective / Orthographic Camera, Stencil, Basic Input, Events Manager, Math module, Loading 3D models with GLTF, Skeletal Meshes, Skeletal Animation, GPU Skinning, Render Graph, Render Passes, PBR Shading, Directional Light, Shadow Mapping, Materials system, Custom Smart Pointers

JavaScript + WebGL 2D Game Engine Jul 2013 - May 2017 (3y 11m)

- Lightweight Web 2D engine. JavaScript, WebGL, GLSL. GitHub Repo
- **Features**: 2D, Texture Atlases, Animation Spritesheets, Scene management, Orthographic Camera, Basic Input

Game AI Project | Pathfinding A*, Flocking, State Machines Apr 2016 - Jul 2016 (4m)

- Java Implementation of autonomous agent behaviors and navigation algorithms (Tiled Map, Pathfinding A*, Flocking, State Machines, Steering Behaviours). GitHub Repo

Transpiler - C++ to Custom OOP Language Oct 2022 - Present

- Transpiler. C++ to Custom OOP Language (Lexer, Parser, AST, Semantic Analysis, Code Generation). GitHub Repo

rss-to-md and md-to-html Python tools Jun 2025

- **rss-to-md**: Download RSS info in Markdown format. GitHub Repo
- **md-to-html**: Convert Markdown to navigable HTML. GitHub Repo

EDUCATION AND LANGUAGES

- Bachelor's Degree in Computer Science and Engineering, University of Murcia - FIUM, 2012 - 2017
- English (Professional)
- Spanish (Native)
- French (Basic)